
AN ENHANCED SOURCE-FREE UNSUPERVISED DOMAIN ADAPTATION FRAMEWORK FOR CROSS-DATASET EEG EMOTION RECOGNITION VIA PREDICTIVE CODING AND TEST-TIME TRAINING

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ABSTRACT

EEG-based emotion recognition is widely used in affective computing but suffers from poor generalization due to domain shifts caused by inter-subject variability, dataset differences, and recording conditions, especially in cross-dataset settings. Conventional unsupervised domain adaptation methods require source data, which is often unavailable due to privacy constraints. Although source-free UDA addresses this limitation, existing methods still struggle with large domain gaps, noisy pseudo-labels, and unstable adaptation. To address these challenges, we propose an enhanced source-free unsupervised domain adaptation (SF-UDA) framework for cross-dataset EEG emotion recognition. The framework introduces a non-contrastive predictive coding-based self-supervised pretraining strategy to learn robust and transferable EEG representations by modeling temporal dependencies in a reconstruction-based manner. During adaptation, we estimate target-domain structure through class-wise clustering and prediction disagreement, and optimize the model using a dual-stage adaptation strategy consisting of Multi-Loss Adaptive Regularization and Localized Consistency Learning. This dual-stage design improves stability and neighborhood consistency under noisy pseudo-labels. Furthermore, we propose a lightweight test-time training mechanism that enables selective online updates for uncertain samples using predictive reconstruction loss and entropy minimization, enhancing robustness during inference while maintaining computational efficiency. Experiments on the DEAP, SEED, and DREAMER datasets show consistent improvements over state-of-the-art SF-UDA methods for both binary and multi-class classification, achieving 69.56% and 63.03% accuracy on SEED and DREAMER when trained on DEAP, and 61.38% and 68.90% when trained on SEED. These results demonstrate the effectiveness of combining predictive self-supervised learning, source-free domain adaptation, and test-time refinement for robust EEG emotion recognition under severe domain shifts.

Keywords EEG emotion recognition · source-free domain adaptation · self-supervised learning · predictive coding · consistency learning

1 Introduction

Emotion is a fundamental aspect of human cognition and behavior, influencing how people think, communicate, and interact. Consequently, emotion recognition has become an important research area in psychology, healthcare, and human–computer interaction. In healthcare, it can assist in diagnosing and monitoring neurological and mental health conditions such as autism spectrum disorder (ASD), depression, and stress-related disorders [1, 2]. It also plays a key role in affective brain–computer interface (aBCI) systems that adapt to users’ emotional states.

Among various modalities, electroencephalogram (EEG) signals have attracted significant attention because they directly reflect brain activity and are less affected by intentional expression or environmental factors than facial expressions or speech [3]. Their high temporal resolution and the increasing availability of wearable EEG devices further support their use in affective computing applications.

Recent advances in deep learning have substantially improved EEG emotion recognition through automatic feature learning from raw or transformed EEG signals [4, 5, 6]. Although convolutional, recurrent, and transformer-based models achieve strong performance within a dataset, their generalization ability often deteriorates across datasets due to domain shifts caused by differences in subjects, recording devices, electrode configurations, and experimental protocols.

To address domain discrepancies, unsupervised domain adaptation (UDA) has been widely studied in EEG emotion recognition [7, 8, 9, 10]. However, most existing methods focus on cross-subject or cross-session adaptation within the same dataset and require access to source-domain data during adaptation. This assumption is often impractical in real-world healthcare and brain–computer interface applications, where sharing raw EEG recordings may be restricted due to privacy, storage, ethical, and legal considerations.

Source-free unsupervised domain adaptation (SF-UDA) has recently emerged as a promising alternative. SF-UDA adapts a model to an unlabeled target domain using only a pre-trained source model, eliminating the need for source-domain data [11]. While this setting is better suited to privacy-sensitive applications, it introduces challenges such as source-model bias, the lack of explicit source-target alignment, and error accumulation from unreliable pseudo-labels.

Although SF-UDA has achieved encouraging results in computer vision [13, 12, 14, 15, 16, 17], its application to EEG analysis remains limited. Existing EEG-related SF-UDA studies mainly focus on seizure and epilepsy detection and sleep stage classification [19, 18, 20], while research on EEG-based emotion recognition is still limited. Moreover, many existing methods rely on single-stage adaptation strategies that struggle to handle the substantial domain discrepancies encountered in cross-dataset EEG emotion recognition.

To address these challenges, we propose an enhanced source-free unsupervised domain adaptation framework for cross-dataset EEG-based emotion recognition, improving upon our previous SF-UDA work [21]. While the earlier version introduced a basic SF-UDA pipeline, the current approach incorporates several key improvements to strengthen target-domain generalization and reduce the effect of noisy pseudo-labels.

The framework consists of four stages: pre-training, computation, target adaptation, and inference. Unlike our earlier work, which relied on supervised source-domain training, we introduce a non-contrastive predictive coding–based self-supervised strategy to learn robust and transferable EEG representations from signals. A predictive coding model is trained to reconstruct future representations from past activity without using negative pairs or contrastive learning, using an encoder–decoder architecture optimized via a predictive reconstruction loss. This design captures temporal dependencies, encourages domain-invariant features, and removes reliance on labeled source data after pretraining, making it suitable for privacy-sensitive settings.

In the computation stage, we estimate the class-wise structure in the target domain using pretrained predictions to derive cluster centroids and classifier disagreement for adaptation. The target adaptation stage consists of two steps using unlabeled data: Multi-Loss Adaptive Regularization (MLAR), which reduces classifier disagreement on high-confidence samples while aligning pseudo-labels to stabilize decision boundaries, and Localized Consistency Learning (LCL), which enforces prediction consistency among reliable neighbors selected using both feature- and classifier-space similarity, reducing the influence of unreliable neighbors.

A key improvement is introduced at inference time. Unlike our previous PC-TTA method [21], we introduce a lightweight test-time training strategy. During inference, samples are evaluated using prediction entropy and predictive error from the temporal predictive coding objective. Uncertain samples are selectively updated with a few gradient steps using predictive and entropy losses, enabling dynamic adaptation and improving robustness under domain shift while avoiding unnecessary computation for confident predictions.

We evaluate the framework on DEAP [22], SEED [23], and DREAMER [24] in a cross-dataset setting. We compare against recent SF-UDA methods [18, 19, 12, 13], and the results show consistent improvements across both binary and multi-class emotion classification tasks in all cross-dataset scenarios.

The main contributions of this work are as follows:

1. We propose an enhanced source-free unsupervised domain adaptation framework for cross-dataset EEG-based emotion recognition that addresses domain shifts and the absence of source data.
2. We introduce a non-contrastive predictive coding-based self-supervised pretraining strategy to learn robust temporal EEG representations without relying on negative pairs or contrastive objectives.
3. We design a lightweight test-time training mechanism to enhance robustness under distribution shift, where uncertain target samples are refined online using predictive reconstruction and entropy minimization.
4. Extensive experiments demonstrate that the multi-stage framework achieves consistent improvements over state-of-the-art SF-UDA baselines in cross-dataset EEG emotion recognition, validating the combined benefit of domain-invariant representation learning and noise-robust pseudo-label optimization.

The remainder of this paper is organized as follows. Section 2 reviews related work on EEG emotion recognition, domain adaptation, and source-free learning. Section 3 presents the proposed framework, including pretraining, adaptation, and test-time training. Section 4 describes the experimental setup, datasets, and results. Section 5 concludes the paper and outlines future directions.

2 Related Work

Extensive research has explored EEG-based emotion recognition using both traditional and deep learning methods. Classical classifiers such as Support Vector Machines, K-Nearest Neighbors, Decision Trees, Logistic Regression, and Linear Discriminant Analysis [25, 26, 27, 28, 29] are widely used but struggle with the non-stationary and temporal nature of EEG signals, limiting generalization to unseen subjects. In deep learning, models such as Autoencoders, Graph Neural Networks, and Deep Belief Networks have been investigated [30, 31, 32], while Convolutional Neural Networks and Long Short-Term Memory networks remain popular for capturing spatial-temporal EEG patterns [34, 33, 4, 6].

Deep learning models require large labeled datasets, which are costly and time-consuming to obtain. Self-supervised learning (SSL) reduces this dependency by learning from unlabeled data through pretext tasks [35, 36]. In EEG, SSL includes contrastive, autoencoder-based, graph-based, and hybrid approaches [37, 38]. While contrastive methods are effective, they depend on negative sampling and large batch sizes. Predictive coding-based methods [39, 40] have recently emerged as a simpler alternative by learning representations through latent prediction rather than instance discrimination.

Despite these advances, performance often degrades under distribution shifts across subjects, sessions, and datasets. UDA addresses this by aligning source and target distributions [34, 41]. Existing EEG-UDA methods focus on feature alignment and pseudo-label refinement, including fine- and coarse-grained alignment [7] and distribution matching and confidence filtering [9]. Adversarial approaches further learn domain-invariant features via competition between feature extractors and discriminators [8, 10].

Only a few studies [42, 43, 44] address cross-dataset EEG emotion recognition, which is particularly challenging due to demographic variability and recording inconsistencies. These works explore dictionary learning [43], subspace projection [42], and test-time augmentation [44] strategies to reduce negative transfer.

Traditional UDA assumes access to source data during adaptation, which is often impractical due to privacy, storage, and computational constraints. Source-Free Domain Adaptation (SFDA) addresses this issue by transferring knowledge from a pretrained source model to unlabeled target data. SFDA methods are broadly categorized into data-centric and model-centric approaches [45]. Data-centric methods enhance target-domain structure and representation through clustering, pseudo-source construction, data augmentation, or iterative pseudo-label refinement [14, 15, 13, 16, 46]. In contrast, model-centric methods directly adapt model on target data using pseudo-labeling, information maximization, and entropy minimization to improve decision boundaries and transferability [12, 11, 47, 48, 17].

Although most SFDA research focuses on computer vision applications [13, 12, 14, 15, 16, 17], its application to time-series signals such as EEG remains limited [18, 19, 20]. Ragab et al. [18] proposed a framework using random masking and temporal imputation to align target and source representations while preserving temporal coherence. Zhao and Peng [19] introduced SS-TrBoosting, a boosting-based SFDA method for EEG seizure classification, later

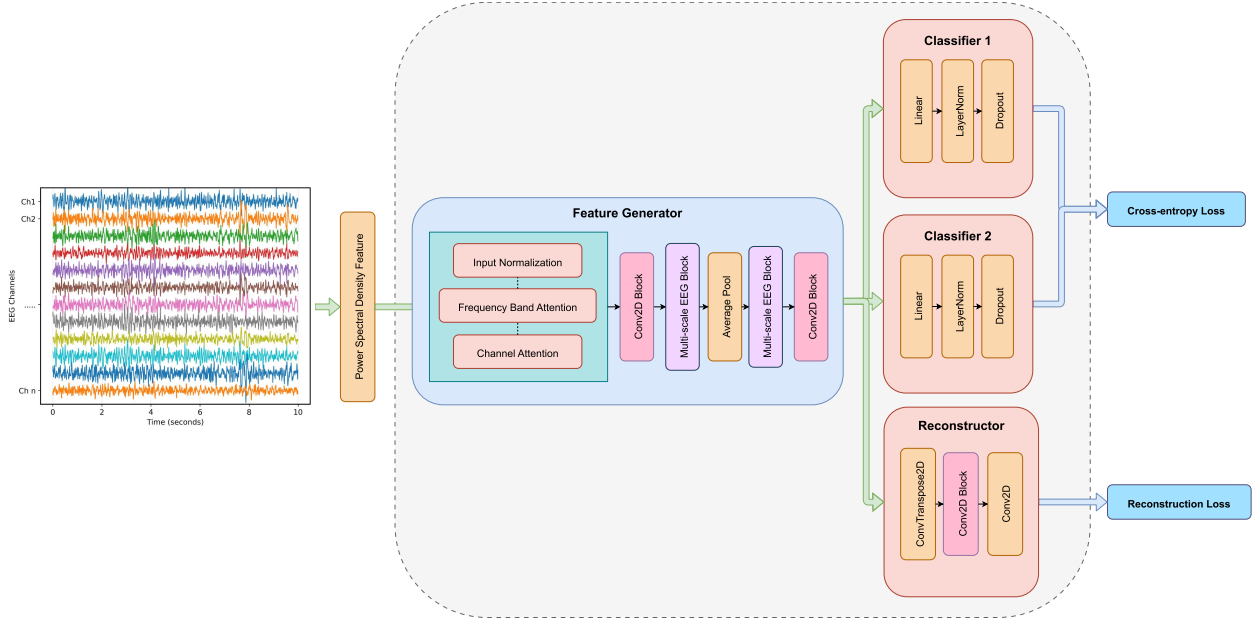


Figure 1: Overall architecture of the proposed network. The feature generator integrates learnable input normalization, axis-wise attention gating, Conv2D blocks (convolution, group normalization, and dropout) for spatial feature learning, and multiscale EEG blocks (stacked convolutions, normalization, adaptive pooling, and dropout) for multiscale EEG representation learning. A reconstructor enables predictive coding via input reconstruction, and two parallel classifiers produce the final prediction by averaging their outputs.

extended to an unsupervised variant (U-TrBoosting). Wang et al. [20] proposed an iEEG-based epilepsy approach that selects informative samples using neighborhood uncertainty and diversity for annotation.

Although deep learning has shown strong performance in EEG-based emotion recognition, cross-domain adaptation remains challenging due to significant distribution shifts between datasets. This is further amplified in source-free settings, where source data is unavailable and conventional alignment methods cannot be applied. Despite its potential, SFDA has been underexplored for emotion recognition. To address this gap, we propose a source-free unsupervised domain adaptation framework for EEG-based emotion classification that handles cross-dataset variability and the absence of source data during adaptation.

3 Proposed Method

Our method consists of four stages: pre-training, computation, target adaptation, and inference. Building on the pipeline of our previous SF-UDA model [21], we introduce key modifications to the pre-training, target adaptation, and inference stages.

3.1 Framework

Let the training dataset be the *source domain* and the testing dataset the *target domain*. The source domain contains N_s labeled EEG samples $X_s = \{x_s^i\}_{i=1}^{N_s}$ with labels $Y_s = \{y_s^i\}_{i=1}^{N_s}$, while the target domain contains N_t unlabeled samples $X_t = \{x_t^i\}_{i=1}^{N_t}$, where x_s^i and x_t^i are the feature representations of the i^{th} EEG segment. The two domains follow different marginal distributions, $P_s(X_s) \neq P_t(X_t)$. We address source-free domain adaptation, where source data is unavailable during adaptation for privacy preservation. The objective is to learn a mapping function f that enables accurate emotion prediction on the target domain by minimizing domain discrepancy through the adaptation stages described below.

The proposed network architecture is illustrated in Fig. 1. The feature generator (F) applies learnable input normalization and axis-wise attention gating, followed by Conv2D and multiscale convolutional blocks with progressive spatial downsampling to extract hierarchical EEG features. The Conv2D block comprises convolution, group normalization,

and dropout, while the multiscale EEG block combines stacked convolutions, normalization, adaptive pooling, and dropout to capture multi-scale representations.

The reconstructor (R), used only during pre-training and test-time training, reconstructs the input via transposed convolutions to support predictive coding and preserve informative temporal structure. Two parallel classifiers (C_1 and C_2), attached to the shared feature representation as in our previous works [21, 44, 49], generate the final prediction by averaging their outputs. Their discrepancy is incorporated into the pre-training loss to promote consistency and robust feature learning. Each classifier consists of two fully connected layers with Layer Normalization, SiLU activation, and dropout.

3.1.1 Pre-training

During pre-training, the model is optimized on labeled source data (X_s, Y_s) to learn discriminative and temporally predictive representations. The objective includes three terms: classification loss (L_{cls}), classifier discrepancy loss (L_{dis}), and predictive coding loss (L_{pred}), as defined in Eq. 3. Following [21, 44], L_{cls} uses weighted cross-entropy, while L_{dis} measures the Euclidean distance between the two classifier outputs to enforce consistency.

In addition, motivated by [40] and the temporal continuity of EEG dynamics, this work introduces a predictive coding module to capture the intrinsic evolution of neural activity. Specifically, given an input (x_t), the feature generator (F) maps it into a latent representation, which is subsequently passed to a reconstructor (R) to predict the power spectral density (PSD) representation of the next segment:

$$\hat{F}_{t+1} = R(F(x_t)) \quad (1)$$

The predictive coding loss is defined as the mean squared error between the predicted and ground-truth future PSD features:

$$L_{pred} = \frac{1}{N_b} \sum_{i=1}^{N_b} \left\| \hat{F}_{t+1}^{(i)} - x_{t+1}^{(i)} \right\|_2^2 \quad (2)$$

where x_{t+1} denotes the ground-truth future PSD representation for the i^{th} sample and N_b is the batch size. This objective encourages the model to capture the underlying temporal structure of EEG signals, reflecting their continuous and predictive nature.

Finally, the overall pre-training objective is:

$$L = L_{cls} + \alpha_1 L_{dis} + \alpha_2 L_{pred} \quad (3)$$

where α_1 and α_2 are weighting factors controlling the contribution of each loss term. Joint optimization of these objectives enables the model to learn discriminative, regularized, and temporally predictive EEG representations.

3.1.2 Computation

After pre-training, the trained network is obtained and source data is no longer accessible. Following [21], cluster centroids (CC_t) for target adaptation are computed by feeding unlabeled target data into the pre-trained model. As in [21, 11], the centroid for emotion class k is:

$$CC_t^k = \frac{\sum_{x_t \in X_t} \hat{p}^k(x_t) F(x_t)}{\sum_{x_t \in X_t} \hat{p}^k(x_t)} \quad (4)$$

where $\hat{p}^k(x_t)$ denotes the combined softmax probability of sample x_t produced by the two classifiers for the k^{th} emotion class:

$$\hat{p}^k(x_t) = \text{avg}(\delta^k(C_1(F(x_t))), \delta^k(C_2(F(x_t)))) \quad (5)$$

where δ represents the softmax function. This approach computes centroids by combining feature representations with classifier predictions, weighting each sample by prediction confidence. In contrast, methods using only feature extractor outputs produce less stable centroids, as they ignore decision boundaries and confidence, increasing ambiguity under domain shift.

3.1.3 Target Adaptation

The target adaptation stage comprises two training steps that operate solely on unlabeled target data using a pre-trained network.

In the first step, we introduce a Multi-Loss Adaptive Regularization (MLAR) framework to enhance consistency and robustness on unlabeled target-domain data. For each training batch, we first generate pseudo-labels by assigning each sample to the nearest cluster centroid in the feature space:

$$\hat{y}_t = \arg \min_k D_l(F(x_t), CC_t^k) \quad (6)$$

where D_l denotes the Euclidean distance computed using the L2 norm. We then construct a reliable training mask by selecting confident samples (CX_t) that satisfy multiple consistency criteria, including agreement between cluster-based pseudo-labels and model predictions, and high prediction confidence:

$$CX_t = \{x_t \in X_{t,b} \mid \hat{y}_t^{(cluster)} = \hat{y}_t^{(model)}, \max(\hat{p}(x_t)) \geq \tau_c\} \quad (7)$$

where $X_{t,b}$ denotes target mini-batch samples, $\hat{y}_t^{(cluster)}$ is the pseudo-label from nearest-centroid assignment, $\hat{y}_t^{(model)}$ is the classifier prediction, $\hat{p}(x_t)$ is the combined softmax probability, and τ_c is the confidence threshold for selecting reliable samples.

The primary learning objective is the pseudo-label agreement loss (L_{plal}), computed over the selected reliable samples:

$$L_{plal} = -\frac{1}{|CX_t|} \sum_{x_t \in CX_t} \log \hat{p}_{\hat{y}_t}(x_t) \quad (8)$$

To further improve classifier consistency, we introduce a classifier discrepancy loss (L_{cdl}), computed as the Jensen-Shannon (JS) divergence between the two classifiers:

$$L_{cdl} = \mathbb{E}_{x_t \in CX_t} [JS(\hat{p}_1(x_t) \parallel \hat{p}_2(x_t))] \quad (9)$$

where $\hat{p}_1$ and $\hat{p}_2$ are the softmax outputs of C_1 and C_2 , respectively.

In addition, we apply two auxiliary regularization terms to stabilize training. Prediction entropy minimization:

$$L_{ent} = -\frac{1}{|CX_t|} \sum_{x_t \in CX_t} \sum_{k=1}^K \hat{p}^k(x_t) \log \hat{p}^k(x_t) \quad (10)$$

Target distribution balancing:

$$L_{bal} = \left\| \frac{1}{N_b} \sum_{x_t \in X_{t,b}} \hat{p}(x_t) - \frac{1}{K} \mathbf{1}_K \right\|_2^2 \quad (11)$$

where K denotes the number of classes and $\mathbf{1}_K$ is a K -dimensional vector of ones. These terms prevent prediction collapse and encourage uniform utilization of classes.

Finally, the overall MLAR objective is:

$$L_{mlar} = \lambda_1 L_{plal} + \lambda_2 L_{cdl} + \lambda_3 L_{ent} + \lambda_4 L_{bal} \quad (12)$$

where $\lambda_1, \lambda_2, \lambda_3$, and λ_4 are weighting factors. At the end of each epoch, cluster centroids are updated using features from the current model, ensuring progressive alignment between pseudo-labels and the evolving target feature space.

In the second step, we employ the Localized Consistency Learning (LCL) strategy [21] to enforce local prediction consistency among neighboring target samples. For each sample, reliable neighbors are selected using both feature-space and classifier-output similarities, and the final set is obtained by intersecting the two nearest-neighbor groups. The model is optimized using the localized consistency loss (L_{lcl}), which minimizes prediction discrepancies among these neighbors. The neighbor selection and loss formulation follow [21].

MLAR improves prediction reliability on unlabeled target data by reducing noise and enforcing consistency. LCL further enhances local agreement among reliable neighbors by aligning feature and classifier spaces.

3.1.4 Inference

We propose uncertainty-based test-time training (Fig. 2) on target samples during inference to mitigate incorrect predictions while improving computational efficiency. During test-time training, the model is updated online using unlabeled target data. To ensure stability, unreliable samples are identified based on predictive uncertainty. For each x_t ,

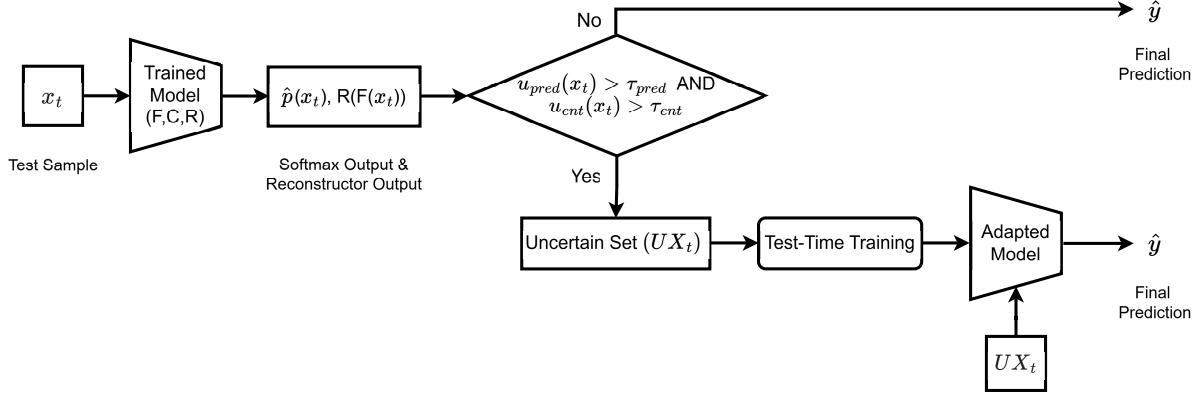


Figure 2: Proposed uncertainty-based test-time training process.

we compute reconstruction error and prediction entropy:

$$u_{pred}(x_t) = \|R(F(x_t)) - x_t\|_2^2 \quad (13)$$

$$u_{ent}(x_t) = - \sum_{k=1}^K \hat{p}^k(x_t) \log \hat{p}^k(x_t) \quad (14)$$

Thresholds are defined using the quantile statistics of the current batch:

$$\tau_{pred} = \text{Quantile}(u_{pred}(X_t), q), \quad \tau_{ent} = \text{Quantile}(u_{ent}(X_t), q) \quad (15)$$

where $q = 0.5$ in our implementation. Based on these thresholds, we construct the uncertain target subset UX_t :

$$UX_t = \{x_t \in X_t \mid u_{pred}(x_t) > \tau_{pred} \wedge u_{ent}(x_t) > \tau_{ent}\} \quad (16)$$

Only selected samples UX_t are used for test-time adaptation, reducing the impact of incorrect predictions and improving computational efficiency. Samples that do not satisfy the uncertainty criterion are excluded, and their predictions remain unchanged. On UX_t , only lightweight components (normalization layers and attention gate modules) are updated using entropy and reconstruction-based objectives. Adaptation is performed for a limited number of iterations to ensure computational efficiency.

Entropy minimization loss:

$$L_{tent} = -\frac{1}{|UX_t|} \sum_{x_t \in UX_t} \sum_{k=1}^K \hat{p}^k(x_t) \log \hat{p}^k(x_t) \quad (17)$$

Predictive reconstruction loss (unlike Eq. 2, which predicts the *next* segment’s PSD during pre-training, this loss reconstructs the *current* sample to measure uncertainty at test time):

$$L_{tppred} = \frac{1}{|UX_t|} \sum_{x_t \in UX_t} \|R(F(x_t)) - x_t\|_2^2 \quad (18)$$

Finally, the overall test-time training objective is:

$$L_{ttt} = \beta_1 L_{tent} + \beta_2 L_{tppred} \quad (19)$$

where β_1 and β_2 are weights for each loss term. Following test-time training with objective L_{ttt} , final predictions for uncertain samples are obtained using the adapted model. Algorithm 1 summarizes the complete procedure.

3.2 Data Preprocessing and Model Input Construction

We use Power Spectral Density (PSD) and Differential Entropy (DE) as EEG features, consistent with prior studies [42, 50]. Although both are evaluated, PSD consistently shows better performance and is therefore adopted in this study.

Require: Source PSD samples X_s , Target PSD samples X_t , Source labels Y_s ; Feature generator F , classifiers C_1, C_2 , reconstructor R ; Epochs: $Epoch_{mlar}, Epoch_{lcl}$

Ensure: Pre-training: Train on (X_s, Y_s) : minimize L (Eq. 3), comprising L_{cls}, L_{dis} , and L_{pred} (Eq. 2)

Computation: Compute cluster centroids CC_t (Eq. 4)

Target Adaptation: for $i = 1$ to $Epoch_{mlar}$ do

- Generate pseudo-labels (Eq. 6); construct CX_t (Eq. 7)
- Minimize L_{mlar} (Eq. 12) on X_t

end

for $i = 1$ to $Epoch_{lcl}$ **do**

- Identify reliable neighbors; minimize L_{lcl} on X_t

end

Inference: for each mini-batch X_t **do**

- Compute u_{pred} (Eq. 13), u_{ent} (Eq. 14), thresholds (Eq. 15), UX_t (Eq. 16)
- Predict certain samples ($X_t \setminus UX_t$) without adaptation
- if** $UX_t \neq \emptyset$ **then**
 - Update normalization layers and attention gates by minimizing L_{ttt} (Eq. 19) on UX_t
 - Predict samples in UX_t using the adapted model

end

end

Algorithm 1: Proposed SF-UDA approach for EEG emotion recognition.

The preprocessing is the same as in our previous methods [21, 44]. Signals are extracted from 32 common channels shared by the DEAP and SEED datasets to ensure consistent input dimensions. Each trial is segmented into 2-second windows with a 1-second overlap, and PSD features are computed across delta (1–3 Hz), theta (4–7 Hz), alpha (8–13 Hz), beta (14–30 Hz), and gamma (31–50 Hz) bands. The resulting features are organized into a 2-D matrix $X \in \mathbb{R}^{n \times i}$, where $n = 32$ and $i = 5$, yielding a 32×5 representation. For the DREAMER dataset, 14 channels are used, producing a 14×5 representation. Finally, all features are normalized to the range $[-1, 1]$.

4 Experiments

4.1 Experimental Details

We evaluate the proposed model on the publicly available DEAP, SEED, and DREAMER datasets, which are widely used in emotion recognition. Cross-dataset experiments are performed by training on one dataset and testing on another.

DEAP [22]. The DEAP dataset involves 32 participants who viewed 40 one-minute music videos. Each trial lasted 63 seconds, including a 3-second pre-trial interval followed by a 60-second video presentation. After each trial, participants rated arousal, valence, dominance, and liking on a 1–9 scale. EEG was recorded using 32 electrodes at 512 Hz, then downsampled to 128 Hz and filtered with a 4–45 Hz band-pass filter.

SEED [23]. The SEED dataset consists of EEG recordings from 15 participants across three sessions held on different days. In each session, participants viewed 15 Chinese film clips designed to elicit positive, neutral, and negative emotions. EEG signals were recorded using 62-channel electrodes at 1,000 Hz, then down-sampled to 200 Hz and band-pass filtered between 0–75 Hz.

DREAMER [24]. The DREAMER dataset contains EEG recordings from 23 participants who watched 18 film clips of varying durations (65–393 seconds). After each clip, participants rated valence, arousal, and dominance on a 1–5 scale. EEG data were collected using 14 electrodes at 128 Hz.

We follow the preprocessing protocol of our previous studies [21, 44]. All subjects and trials are used. In DEAP, the first 3 seconds of each trial are removed. SEED and DREAMER signals are band-pass filtered between 0.3–50 Hz [51, 21, 44], while DEAP is kept unchanged as it is already filtered (4–45 Hz). We perform both binary and three-class emotion recognition. For binary classification, DEAP uses a valence threshold of 4.5, DREAMER uses 3.0 [42, 52, 21], and SEED includes only positive and negative classes. For multi-class classification, DEAP is split into negative (1–4), neutral (4–6), and positive (6–9) [53, 21], SEED retains its original labels, and DREAMER is labeled as negative (1–2), neutral (3), and positive (4–5).

Table 1: Binary emotion classification accuracy comparison with state-of-the-art SF-UDA methods. Statistical significance is indicated using a paired-sample t -test: $\sim$ *nonsignificant*, $*p < 0.05$, $**p < 0.01$.

Method	Accuracy (%)	
	SEED→DEAP	DEAP→SEED
Ahmed et al. [12]	51.71**	61.33**
Du et al. [13]	48.74**	56.45**
Ragab et al. [18]	53.55**	59.57**
Zhao et al. [19]	53.17**	61.98**
Our previous SF-UDA [21]	58.99*	65.84*
Proposed method	61.38	69.56

Table 2: Multiclass emotion (positive, neutral, negative) classification accuracy. Significance notation same as Table 1.

Method	Accuracy (%)	
	SEED→DEAP	DEAP→SEED
Ahmed et al. [12]	44.08**	51.29**
Du et al. [13]	39.93**	47.35**
Ragab et al. [18]	45.97**	51.82**
Zhao et al. [19]	47.45**	53.60*
Our previous SF-UDA [21]	51.50*	57.37
Proposed method	53.87	57.08

Table 3: Emotion classification accuracy comparison with other SF-UDA methods for binary and three-class settings, where DREAMER is used as the target dataset. Significance notation same as Table 1.

Method	Binary Emotion (%)		Three-Class Emotion (%)	
	SEED→DREAMER	DEAP→DREAMER	SEED→DREAMER	DEAP→DREAMER
Ahmed et al. [12]	51.47**	44.08**	46.39**	37.28**
Du et al. [13]	55.65**	54.46**	48.65**	49.00*
Ragab et al. [18]	58.26**	54.90**	49.02**	49.15*
Zhao et al. [19]	60.29**	53.75**	55.96**	50.80*
Our previous SF-UDA [21]	67.08	58.87**	59.44*	53.83*
Proposed method	68.90	63.03	62.67	56.24

All experiments are conducted on a Linux system using Python 3.12.13 and PyTorch 2.11.0+cu128, with an NVIDIA Tesla T4 GPU (15 GB). The batch size is 64. The hyperparameters α_1 , α_2 , λ_1 , λ_2 , λ_3 , λ_4 , β_1 , β_2 , and τ_c are set to 0.5, 0.5, 0.5, 0.5, 0.1, 0.1, 0.1, 0.5, and 0.8 respectively.

4.2 Results and Discussion

We evaluate performance using cross-dataset testing by training on one dataset and testing on another in both directions. Our method is compared with our previous SF-UDA approach [21]. Since no other SF-UDA methods have been specifically designed for EEG emotion recognition, we further compare against four general-purpose state-of-the-art SF-UDA methods [18, 19, 12, 13]. All open-source baselines are reimplemented and evaluated on the same datasets to ensure a fair comparison.

Table 1 presents the binary cross-dataset emotion classification results. The proposed method consistently achieves the best performance in both transfer directions, reaching 61.38% for SEED → DEAP and 69.56% for DEAP → SEED, outperforming all competing SF-UDA methods, including our previous approach [21]. Paired-sample t -tests confirm that improvements are statistically significant (*) compared with our previous method and highly significant (**) compared with the remaining baselines.

We further evaluate the model on the three-class emotion recognition task (Table 2). The proposed method achieves significantly better performance on SEED → DEAP compared to all competing approaches. For DEAP → SEED, our previous SF-UDA method [21] performs slightly better, while the proposed method remains highly competitive.

To further assess generalization, additional experiments are conducted using DREAMER as the target dataset. As shown in Table 3, the proposed method achieves the best performance across all adaptation settings. For binary classification, it attains 68.90% for SEED→DREAMER and 63.03% for DEAP→DREAMER, surpassing our previous SF-UDA method by 1.82% and 4.16%, respectively. In the three-class setting, it also achieves the highest accuracies of 62.67% and 56.24%.

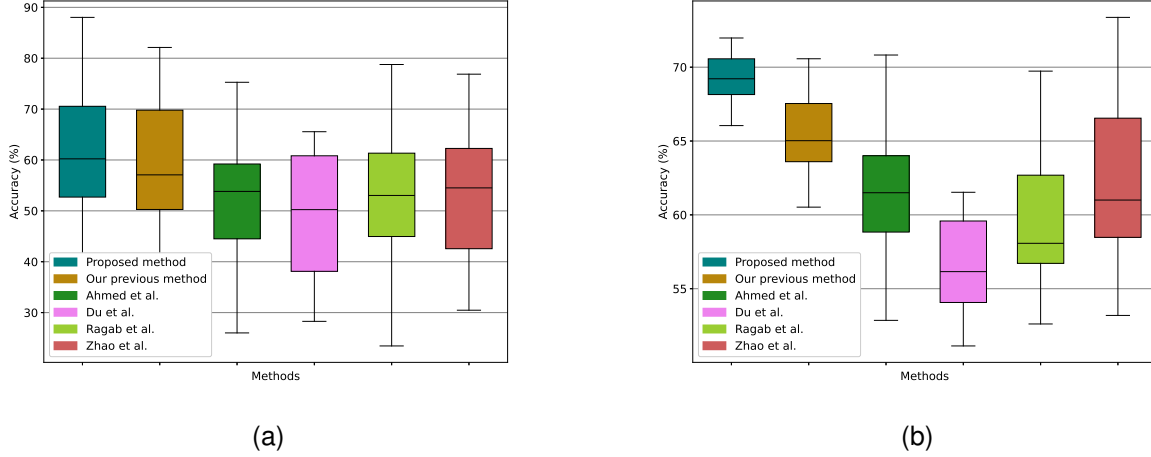


Figure 3: Subject-level accuracy distributions using boxplots under cross-dataset conditions: (a) SEED $\rightarrow$ DEAP and (b) DEAP $\rightarrow$ SEED. The proposed model achieves the highest median accuracy in both cases while exhibiting lower variability across subjects.

Table 4: Performance comparison (PPV, sensitivity, and F1) for positive and negative emotion identification.

Method	SEED $\rightarrow$ DEAP						DEAP $\rightarrow$ SEED					
	Negative			Positive			Negative			Positive		
	PPV	Se	F1	PPV	Se	F1	PPV	Se	F1	PPV	Se	F1
Ahmed et al. [12]	32.67	29.11	30.79	61.04	64.93	62.92	66.44	45.73	54.17	58.65	76.91	66.55
Du et al. [13]	26.72	22.35	24.34	58.57	64.17	61.24	59.00	42.18	49.19	55.03	70.70	61.89
Ragab et al. [18]	33.38	26.01	29.24	61.69	69.66	65.43	62.15	48.87	54.72	57.90	70.26	63.48
Zhao et al. [19]	35.31	32.36	33.77	62.30	65.34	63.78	64.93	52.04	57.77	60.01	71.91	65.42
Our previous [21]	41.59	27.57	33.16	64.63	77.37	70.43	72.70	50.69	59.73	62.17	80.97	70.34
Proposed	46.53	31.40	37.50	66.30	78.91	72.06	71.65	64.71	68.00	67.85	74.41	70.98

Fig. 3 shows boxplots of subject-wise accuracy distributions. For DEAP $\rightarrow$ SEED, it achieves a median accuracy of 69.21% with noticeably lower variability than other methods. In SEED $\rightarrow$ DEAP, it obtains a median accuracy of 60.22%, again outperforming all baselines. Overall, the median accuracy improves by 4.19% and 3.15% for DEAP $\rightarrow$ SEED and SEED $\rightarrow$ DEAP, respectively, over our previous method [21].

Table 4 and Fig. 4 compare the performance of the proposed method with existing SF-UDA approaches for binary emotion recognition. The proposed framework achieves the highest F1-scores across both transfer settings, reaching 37.50% (negative) and 72.06% (positive) for SEED $\rightarrow$ DEAP, and 68.00% (negative) and 70.98% (positive) for DEAP $\rightarrow$ SEED. The confusion matrices further compare the proposed method with the best-performing competing SF-UDA approaches, demonstrating fewer misclassifications and more consistent class-wise predictions. The lower performance on DEAP may be partly attributable to its class imbalance, whereas SEED is more balanced.

To qualitatively evaluate the adaptation process, we visualize the learned feature representations using t-SNE for the SEED $\rightarrow$ DEAP and DEAP $\rightarrow$ SEED settings (Fig. 5). Before adaptation, features from different emotion classes exhibit substantial overlap. After adaptation, the features become more compact and better separated. Although perfect separation is not achieved, the improved clustering pattern indicates enhanced feature alignment and class discriminability.

4.2.1 Ablation study

We perform an ablation analysis to investigate the effectiveness of the major components. Three ablated variants are considered: Model A excludes the complete adaptation process; Model B removes the MLAR module; Model C disables test-time training during inference while preserving all other components.

In addition to evaluating models using PSD features, we further examine their performance with DE features (Table 5). Overall, PSD achieves better performance, except for a slight advantage of DE for Model C in the DEAP $\rightarrow$ SEED

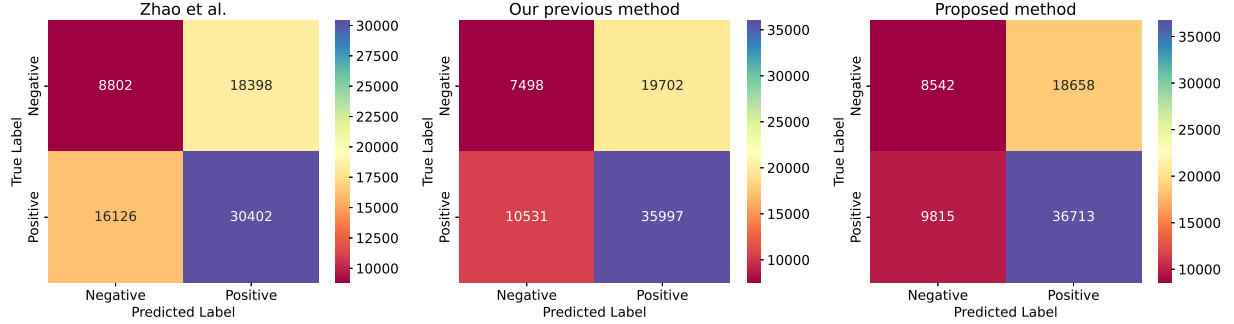


Figure 4: Confusion matrices comparing the proposed method with the strongest SF-UDA baselines on SEED → DEAP, demonstrating improved class-wise consistency and reduced inter-class confusion.

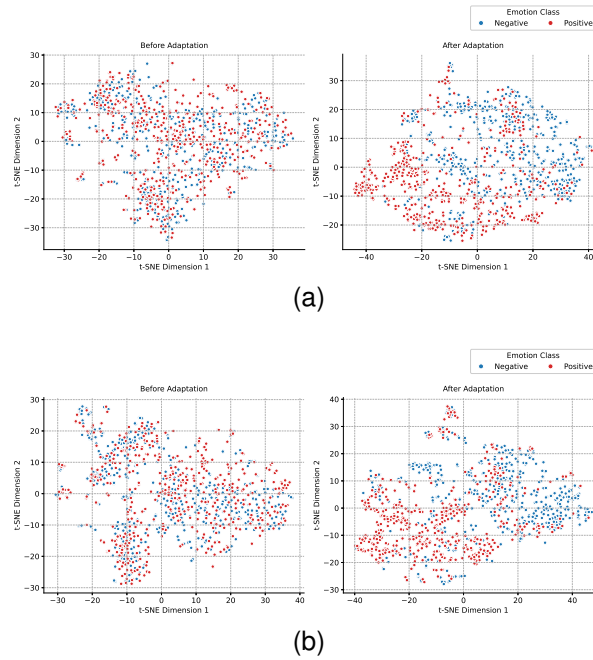


Figure 5: t-SNE visualization of feature distributions before and after adaptation: (a) SEED → DEAP and (b) DEAP → SEED. The proposed method produces more discriminative and well-separated embeddings after adaptation.

Table 5: Overall accuracy comparison in the ablation study (PSD and DE features).

Model	SEED→DEAP (%)		DEAP→SEED (%)	
	PSD	DE	PSD	DE
Model A	51.28	51.11	57.00	56.80
Model B	57.09	55.34	63.96	62.49
Model C	57.33	57.27	64.15	64.68
Proposed	61.38	59.93	69.56	69.14

setting. PSD is therefore adopted for the final framework. Model A consistently yields the lowest performance, highlighting the importance of target-domain adaptation. Removing MLAR (Model B) reduces accuracy by 4.29% and 5.60% for SEED → DEAP and DEAP → SEED, respectively (PSD). Disabling test-time training (Model C) decreases accuracy by 4.05% and 5.41%. These findings confirm the significant contributions of both modules.

The results in Table 6 illustrate the impact of different uncertainty-based gating strategies within the TTT framework. Combining entropy with predictive error yields the most consistent and best performance, suggesting that predictive error provides more reliable sample-level guidance than discrepancy when paired with entropy.

Table 6: Effect of different uncertainty-based gating strategies in test-time training.

Uncertainty selection strategy	Accuracy (%)	
	SEED→DEAP	DEAP→SEED
Entropy + Discrepancy	59.74	68.10
Entropy + Predictive error	61.38	69.56
Discrepancy + Predictive error	60.44	68.88
All three criteria combined	60.35	69.27

Table 7: Ablation studies of Predictive Coding (PredCod) integration and test-time training (TTT) components.

(a) PredCod integration strategy				
PredCod integration strategy	Accuracy (%)			
	SEED→DEAP	DEAP→SEED		
No PredCod	56.30	62.37		
PredCod in pre-training only	59.68	67.41		
PredCod in TTT only	57.39	66.06		
PredCod in both (proposed)	61.38	69.56		

(b) Model update components in TTT				
TTT update components	SEED→DEAP		DEAP→SEED	
	Acc.(%)	Time(s)	Acc.(%)	Time(s)
No adaptation	57.33	7.62	64.15	11.00
Classifier only	59.05	24.88	65.80	34.51
Normalization layers only	60.21	18.11	68.11	27.53
Attention modules only	58.84	20.36	67.47	30.04
Full model updates	60.13	47.40	69.73	64.72
Norm. + Attention (proposed)	61.38	22.04	69.56	31.28

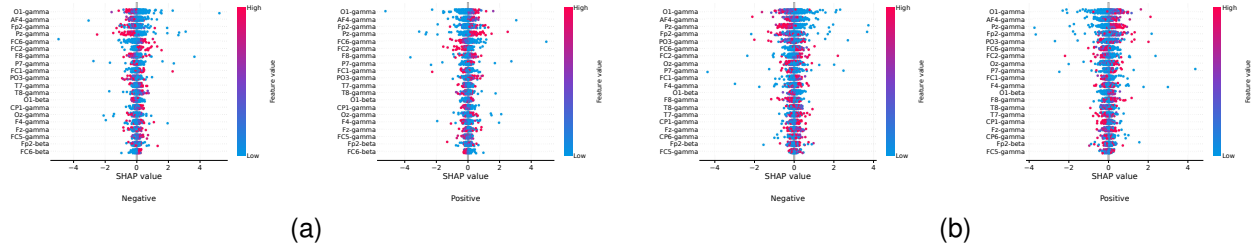


Figure 6: Distributions of SHAP values for representative subjects under cross-dataset settings: (a) SEED → DEAP (Subject 3), and (b) DEAP → SEED (Subject 1). SHAP analysis is performed for positive and negative emotion classes, showing the top 20 EEG features ranked by importance. Features are denoted as channel–frequency band pairs (e.g., O1–gamma). Wider dispersion reflects stronger feature influence; dot color encodes feature magnitude from low (blue) to high (red).

Table 7(a) presents an ablation study on the impact of Predictive Coding (PredCod) during pre-training and TTT. PredCod consistently improves performance in both settings. The best results are obtained when PredCod is used in both stages, highlighting the benefit of modeling intrinsic neural dynamics throughout the learning pipeline.

Table 7(b) evaluates different TTT component update strategies. All TTT variants outperform the non-adaptive baseline, confirming the effectiveness of test-time adaptation. Among partial updates, normalization layers yield more stable gains. Although full model updates slightly outperform our method for DEAP→SEED (by 0.17%), they underperform for SEED→DEAP, primarily due to overfitting on uncertain and potentially noisy test samples. Full model updates are also the most computationally expensive, requiring $2.15\times$ and $2.07\times$ more time than the proposed method for SEED→DEAP and DEAP→SEED, respectively. The proposed selective adaptation achieves a better trade-off between efficiency and performance, reducing inference overhead compared to PC-TTA [21, 44] by avoiding multiple augmented forward passes and using fewer update steps.

4.2.2 EEG feature significance and interpretability analysis

To interpret the model’s predictions and identify influential EEG features, we use SHapley Additive exPlanations (SHAP) [54]. Fig. 6 shows SHAP value distributions for representative subjects. SHAP values are computed for

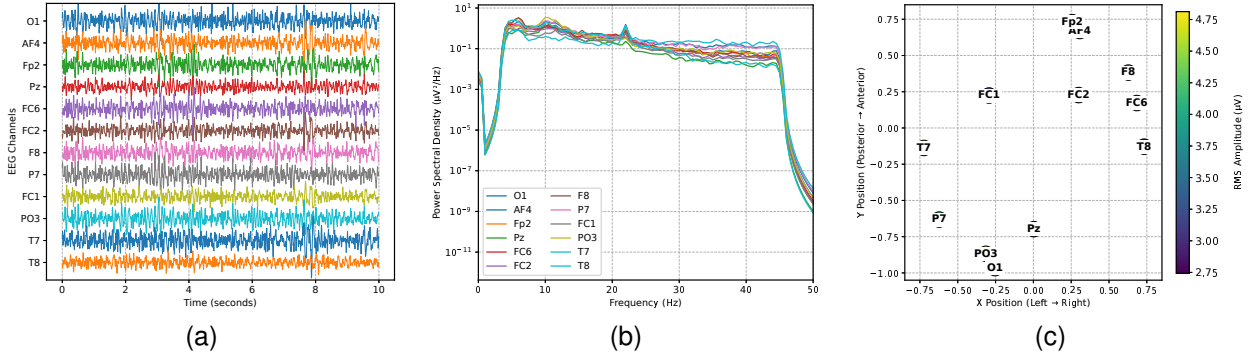


Figure 7: Multi-domain visualization of the top 12 EEG electrodes: (a) temporal signal evolution, (b) frequency-domain PSD profiles, and (c) spatial scalp map. The visualizations jointly characterize the temporal, spectral, and spatial properties of discriminative EEG features.

positive and negative emotions, and the top 20 channel–band pairs (across 32 channels and five frequency bands) are selected based on the mean absolute SHAP values.

The most influential features concentrate in occipital, prefrontal, frontal-central, temporal, and parietal regions, with gamma and beta bands dominating across subjects. A consistent set of influential channels emerges across experimental runs, including O1, AF4, Fp2, Pz, FC6, FC2, FC1, F8, F4, T7, T8, PO3, and Oz. Notably, the occipital channel O1 consistently ranks first. Gamma and beta band features consistently exhibit high SHAP values, underscoring the role of higher-frequency neural oscillations in emotion discrimination. These findings align with prior neurophysiological studies on emotion processing [21, 55, 56, 57].

We further present a multi-domain analysis using the top 12 electrodes (Fig. 7). The temporal view reveals discriminative neural patterns over time; the frequency-domain view presents PSD profiles; and the spatial view maps electrode locations onto a 2D scalp layout. Together, these visualizations provide a unified interpretation of important EEG features in terms of their temporal, spectral, and spatial characteristics.

The results demonstrate that the proposed method achieves strong and consistent performance across multiple datasets. Even under significant distribution shifts and without access to source-domain data, it reliably outperforms existing state-of-the-art approaches.

5 Conclusion

In this work, we propose an enhanced source-free unsupervised domain adaptation framework for cross-dataset EEG-based emotion recognition. The method addresses domain shifts, absence of source data, and unreliable pseudo-labels through a unified multi-stage design. We introduce a non-contrastive predictive coding–based self-supervised pre-training strategy to learn robust EEG representations by modeling temporal dependencies without labeled data. During adaptation, Multi-Loss Adaptive Regularization and Localized Consistency Learning jointly improve prediction stability and neighborhood consistency under noisy supervision. A lightweight test-time training mechanism further refines uncertain samples, enhancing robustness during inference. Extensive experiments on the DEAP, SEED, and DREAMER datasets demonstrate that the proposed approach consistently outperforms state-of-the-art SF-UDA methods in both binary and multi-class settings. Future work will explore continual and lifelong adaptation for evolving target distributions, as well as extensions to other physiological signals such as ECG and EMG, and multimodal EEG fusion for more robust affective computing systems.

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References

- [1] R. Guo, H. Guo, L. Wang, M. Chen, D. Yang, and B. Li, “Development and application of emotion recognition technology—a systematic literature review,” *BMC Psychology*, vol. 12, no. 1, p. 95, 2024.
- [2] M. A. Rashidan et al., “Technology-assisted emotion recognition for autism spectrum disorder (ASD) children: a systematic literature review,” *IEEE Access*, vol. 9, pp. 33638–33653, 2021.
- [3] Y. Wang, B. Zhang, and L. Di, “Research progress of EEG-based emotion recognition: A survey,” *ACM Computing Surveys*, vol. 56, no. 11, pp. 1–49, 2024.
- [4] T. Wang, X. Huang, Z. Xiao, W. Cai, and Y. Tai, “EEG emotion recognition based on differential entropy feature matrix through 2D-CNN-LSTM network,” *EURASIP J. Adv. Signal Process.*, vol. 2024, no. 1, p. 49, 2024.
- [5] N. Kouka, R. Fourati, R. Fdhila, P. Siarry, and A. M. Alimi, “EEG channel selection-based binary particle swarm optimization with recurrent convolutional autoencoder for emotion recognition,” *Biomed. Signal Process. Control*, vol. 84, p. 104783, 2023.
- [6] C. Fan, H. Xie, J. Tao, Y. Li, G. Pei, T. Li, and Z. Lv, “Icaps-reslstm: Improved capsule network and residual lstm for eeg emotion recognition,” *Biomed. Signal Process. Control*, vol. 87, p. 105422, 2024.
- [7] M. Jiménez-Guarneros and G. Fuentes-Pineda, “Learning a robust unified domain adaptation framework for cross-subject EEG-based emotion recognition,” *Biomed. Signal Process. Control*, vol. 86, p. 105138, 2023.
- [8] T. Li, Z. Wang, and H. Liu, “Adversarial domain adaptation-based EEG emotion transfer recognition,” *IEEE Access*, 2025.
- [9] J. Tang et al., “UDA-DDA: Unsupervised domain adaptation with dynamic distribution alignment network for emotion recognition using EEG signals,” *Neurocomputing*, 2025, p. 131715.
- [10] Z. He, Y. Zhong, and J. Pan, “An adversarial discriminative temporal convolutional network for EEG-based cross-domain emotion recognition,” *Comput. Biol. Med.*, vol. 141, p. 105048, 2022.
- [11] J. Liang, D. Hu, and J. Feng, “Do we really need to access the source data? Source hypothesis transfer for unsupervised domain adaptation,” in *Proc. ICML*, 2020, pp. 6028–6039.
- [12] S. Ahmed, A. A. Arafat, M. N. Rizve, R. Hossain, Z. Guo, and A. S. Rakin, “SSDA: Secure source-free domain adaptation,” in *Proc. IEEE/CVF ICCV*, 2023, pp. 19180–19190.
- [13] Y. Du, H. Yang, M. Chen, H. Luo, J. Jiang, Y. Xin, and C. Wang, “Generation, augmentation, and alignment: A pseudo-source domain-based method for source-free domain adaptation,” *Mach. Learn.*, vol. 113, no. 6, pp. 3611–3631, 2024.
- [14] X. Liu, J. Wu, T. Lu, S. Zhang, and G. Wang, “SRPL-SFDA: SAM-guided reliable pseudo-labels for source-free domain adaptation in medical image segmentation,” *Neurocomputing*, vol. 649, p. 130749, 2025.
- [15] J. Jiang, Q. Lv, Y. Li, Y. Du, J. Dong, S. Chen, and H. Yu, “HG-SFDA: HyperGraph learning meets source-free unsupervised domain adaptation,” *IEEE Trans. Image Process.*, vol. 34, pp. 7542–7557, 2025.
- [16] Y. Huang, X. Yang, J. Zhang, and C. Xu, “Relative alignment network for source-free multimodal video domain adaptation,” in *Proc. 30th ACM MM*, 2022, pp. 1652–1660.
- [17] C. Yang, X. Guo, Z. Chen, and Y. Yuan, “Source-free domain adaptation for medical image segmentation with Fourier style mining,” *Med. Image Anal.*, vol. 79, p. 102457, 2022.
- [18] M. Ragab, E. Eldele, M. Wu, C.-S. Foo, X. Li, and Z. Chen, “Source-free domain adaptation with temporal imputation for time series data,” in *Proc. 29th ACM SIGKDD*, 2023, pp. 1989–1998.
- [19] C. Zhao, R. Peng, and D. Wu, “Source-free domain adaptation (SFDA) for privacy-preserving seizure subtype classification,” *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 31, pp. 2315–2325, 2023.
- [20] K. Wang, M. Yang, A. Liu, C. Li, R. Qian, and X. Chen, “Active source-free domain adaptation for intracranial EEG classification via neighborhood uncertainty and diversity,” *Biomed. Signal Process. Control*, vol. 104, p. 107464, 2025.
- [21] M. N. Imtiaz and N. Khan, “Towards practical emotion recognition: An unsupervised source-free approach for EEG domain adaptation,” *arXiv preprint arXiv:2504.03707*, 2025.
- [22] S. Koelstra et al., “DEAP: A database for emotion analysis using physiological signals,” *IEEE Trans. Affective Comput.*, vol. 3, no. 1, pp. 18–31, 2011.
- [23] W.-L. Zheng and B.-L. Lu, “Investigating critical frequency bands and channels for EEG-based emotion recognition with deep neural networks,” *IEEE Trans. Auton. Ment. Dev.*, vol. 7, no. 3, pp. 162–175, 2015.

- [24] S. Katsigiannis and N. Ramzan, “DREAMER: A database for emotion recognition through EEG and ECG signals from wireless low-cost off-the-shelf devices,” *IEEE J. Biomed. Health Informat.*, vol. 22, no. 1, pp. 98–107, 2017.
- [25] Q. Cai, G.-C. Cui, and H.-X. Wang, “EEG-based emotion recognition using multiple kernel learning,” *Mach. Intell. Res.*, vol. 19, no. 5, pp. 472–484, 2022.
- [26] P. Patel, S. Balasubramanian, and R. N. Annavarapu, “Cross subject emotion identification from multichannel EEG sub-bands using Tsallis entropy feature and KNN classifier,” *Brain Informat.*, vol. 11, no. 1, p. 7, 2024.
- [27] Y. Zheng, J. Ding, F. Liu, and D. Wang, “Adaptive neural decision tree for EEG based emotion recognition,” *Inf. Sci.*, vol. 643, p. 119160, 2023.
- [28] A. R. Singh, G. Singh, and N. Saluja, “Emotion classification of EEG signals using logistic regression classification,” in *Proc. 3rd Int. Conf. Innov. Technol. (INOCON)*, 2024, pp. 1–5.
- [29] M. Melinda et al., “Classification of EEG signal using independent component analysis and discrete wavelet transform based on linear discriminant analysis,” *JOIV: Int. J. Informat. Vis.*, vol. 7, no. 3, pp. 830–838, 2023.
- [30] M. Pang, H. Wang, J. Huang, C.-M. Vong, Z. Zeng, and C. Chen, “Multi-scale masked autoencoders for cross-session emotion recognition,” *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 32, pp. 1637–1646, 2024.
- [31] X. Lin, J. Chen, W. Ma, W. Tang, and Y. Wang, “EEG emotion recognition using improved graph neural network with channel selection,” *Comput. Methods Programs Biomed.*, vol. 231, p. 107380, 2023.
- [32] S. Wang, Z. Luo, S. Zhao, Q. Zhang, G. Liu, D. Wu, E. Yin, and C. Chen, “Classification of EEG signals based on sparrow search algorithm-deep belief network for brain-computer interface,” *Bioengineering*, vol. 11, no. 1, p. 30, 2023.
- [33] Z. Shi, Z. Liao, and H. Tabata, “Enhancing performance of convolutional neural network-based epileptic electroencephalogram diagnosis by asymmetric stochastic resonance,” *IEEE J. Biomed. Health Informat.*, vol. 27, no. 9, pp. 4228–4239, 2023.
- [34] C. Xu, Y. Song, Q. Zheng, Q. Wang, and P.-A. Heng, “Unsupervised multi-source domain adaptation via contrastive learning for EEG classification,” *Expert Syst. Appl.*, vol. 261, p. 125452, 2025.
- [35] K. Zhang et al., “Self-supervised learning for time series analysis: Taxonomy, progress, and prospects,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 10, pp. 6775–6794, 2024.
- [36] X. Ren, W. Wei, L. Xia, and C. Huang, “A comprehensive survey on self-supervised learning for recommendation,” *ACM Comput. Surv.*, vol. 58, no. 1, pp. 1–38, 2025.
- [37] X. Wang, Y. Ma, J. Cammon, F. Fang, Y. Gao, and Y. Zhang, “Self-supervised EEG emotion recognition models based on CNN,” *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 31, pp. 1952–1962, 2023.
- [38] K. G. Montero Quispe, D. M. S. Utyiama, E. M. dos Santos, H. A. B. F. Oliveira, and E. J. P. Souto, “Applying self-supervised representation learning for emotion recognition using physiological signals,” *Sensors*, vol. 22, no. 23, p. 9102, 2022.
- [39] M. Sprevak, “Predictive coding I: Introduction,” *Philos. Compass*, vol. 19, no. 1, p. e12950, 2024.
- [40] N. Sheibani and N. Khan, “PC-SSL: A predictive coding-based self-supervised learning framework for EEG emotion recognition,” in *Proc. IEEE ICASSP*, pp. 1676–1680, 2026.
- [41] C. Zhu, L. Zhang, W. Luo, G. Jiang, and Q. Wang, “Tensorial multiview low-rank high-order graph learning for context-enhanced domain adaptation,” *Neural Netw.*, vol. 181, p. 106859, 2025.
- [42] T. Ni, Y. Ni, and J. Xue, “A domain adaptation sparse representation classifier for cross-domain electroencephalogram-based emotion classification,” *Front. Psychol.*, vol. 12, p. 721266, 2021.
- [43] X. Gu, W. Cai, M. Gao, Y. Jiang, X. Ning, and P. Qian, “Multi-source domain transfer discriminative dictionary learning modeling for electroencephalogram-based emotion recognition,” *IEEE Trans. Comput. Soc. Syst.*, vol. 9, no. 6, pp. 1604–1612, 2022.
- [44] M. N. Imtiaz and N. Khan, “Enhanced cross-dataset electroencephalogram-based emotion recognition using unsupervised domain adaptation,” *Comput. Biol. Med.*, vol. 184, p. 109394, 2025.
- [45] J. Li, Z. Yu, Z. Du, L. Zhu, and H. T. Shen, “A comprehensive survey on source-free domain adaptation,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5743–5762, Aug. 2024.
- [46] J. Tian, H. Luo, S. Wu, P. Yan, and J. Zhang, “Source-free domain adaptation for open-set cross-domain fault diagnosis,” *IEEE Trans. Ind. Informat.*, 2025.
- [47] X. Li, Z. Du, J. Li, L. Zhu, and K. Lu, “Source-free active domain adaptation via energy-based locality preserving transfer,” in *Proc. 30th ACM MM*, 2022, pp. 5802–5810.

- [48] Y. Liu, Y. Chen, W. Dai, M. Gou, C.-T. Huang, and H. Xiong, "Source-free domain adaptation with domain generalized pretraining for face anti-spoofing," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5430–5448, 2024.
- [49] M. N. Imtiaz and N. Khan, "Cross-database and cross-channel electrocardiogram arrhythmia heartbeat classification based on unsupervised domain adaptation," *Expert Syst. Appl.*, vol. 244, p. 122960, 2024.
- [50] X. Li, Y. Zhang, P. Tiwari, D. Song, B. Hu, M. Yang, Z. Zhao, N. Kumar, and P. Marttinen, "EEG-based emotion recognition: A tutorial and review," *ACM Comput. Surv.*, vol. 55, no. 4, pp. 1–57, 2022.
- [51] J. Li, S. Qiu, C. Du, Y. Wang, and H. He, "Domain adaptation for EEG emotion recognition based on latent representation similarity," *IEEE Trans. Cogn. Dev. Syst.*, vol. 12, no. 2, pp. 344–353, 2019.
- [52] X. Zhang, X. Cheng, and H. Liu, "TPRO-NET: an EEG-based emotion recognition method reflecting subtle changes in emotion," *Sci. Rep.*, vol. 14, no. 1, p. 13491, 2024.
- [53] S. Tripathi, S. Acharya, R. Sharma, S. Mittal, and S. Bhattacharya, "Using deep and convolutional neural networks for accurate emotion classification on DEAP data," in *Proc. AAAI Conf. Artif. Intell.*, vol. 31, no. 2, pp. 4746–4752, 2017.
- [54] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," *Adv. Neural Inf. Process. Syst.*, vol. 30, 2017.
- [55] R. Nakajima, M. Kinoshita, H. Okita, Z. Liu, and M. Nakada, "Preserving right pre-motor and posterior pre-frontal cortices contribute to maintaining overall basic emotion," *Front. Hum. Neurosci.*, vol. 15, p. 612890, 2021.
- [56] F. I. Mouri, C. E. Valderrama, and S. G. Camorlinga, "Identifying relevant asymmetry features of EEG for emotion processing," *Front. Psychol.*, vol. 14, p. 1217178, 2023.
- [57] K. S. Park, H. Choi, K. J. Lee, J. Y. Lee, K. O. An, and E. J. Kim, "Emotion recognition based on the asymmetric left and right activation," *Int. J. Med. Med. Sci.*, vol. 3, no. 6, pp. 201–209, 2011.